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Objectives

- Implement mechanisms of early-stage cognitive development in virtual agents and robots
- Autonomous construction of spatial awareness
- Autonomous learning of a notion of objects
- Emergence of intelligent behaviours

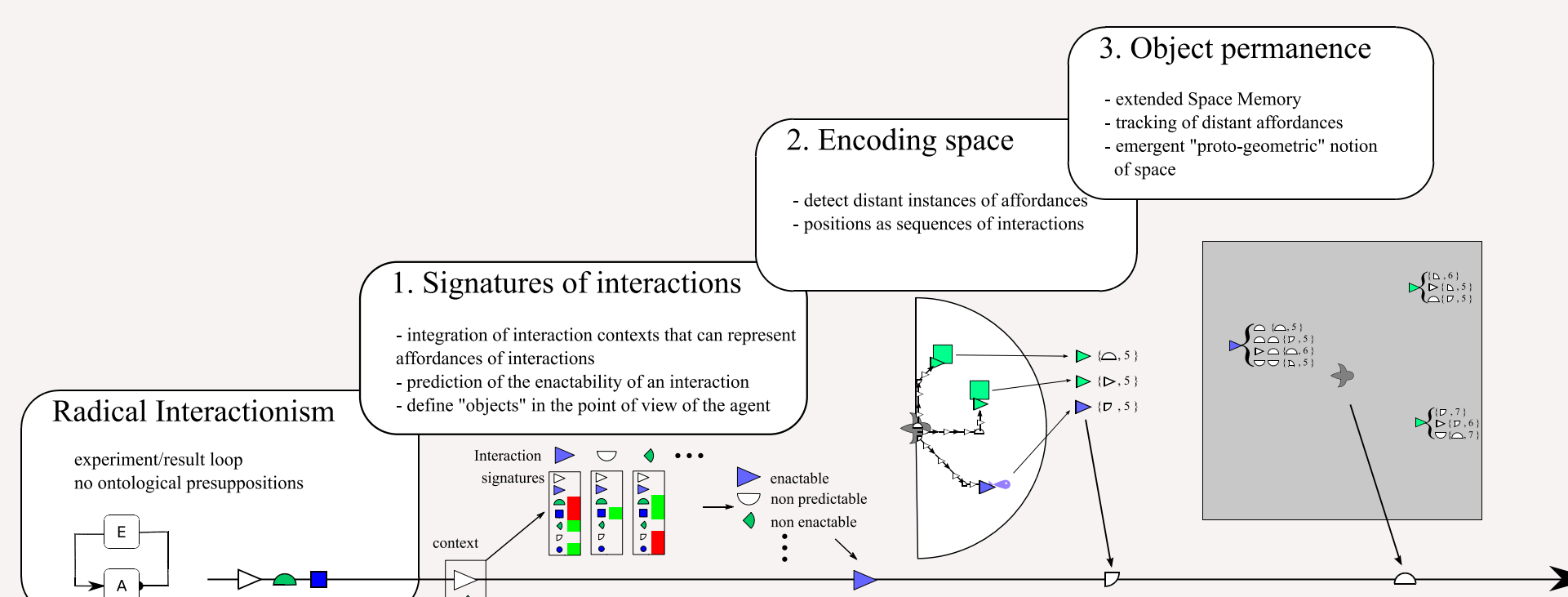
The Radical Interactionism approach [3]:

- An intrinsically motivated mechanism based on interactions
- Inspiration from cognitive theories:
 - Bundle Theory (e.g. Hume, 1739)
 - Representational Space (e.g. Poincaré, 1905)
 - Theory of affordances (e.g. Gibson, 1977)

Space Memory

Principle of the Egocentric Space Memory [4]:

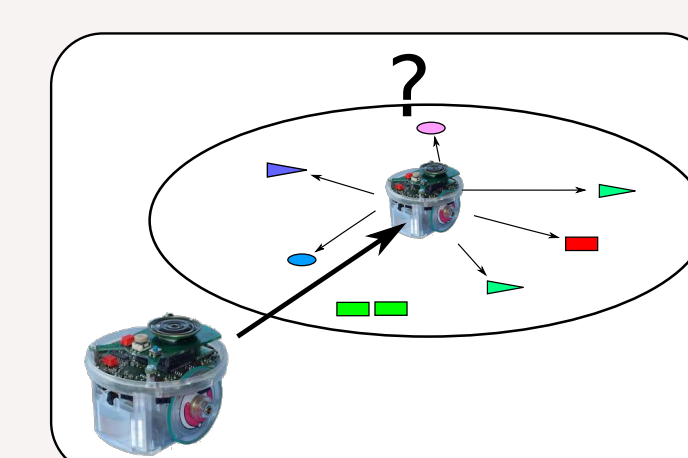
- Learn to define "objects" that afford interactions
- Learn relations between interactions and space



- Short-term egocentric context of surrounding affordances
- Emergence of object permanence
- Enables reference changes

Challenges

- Integration, recognition and localization of mobile elements in the surrounding space [2]
- Prediction of other agents' movements based on inferred behavioral preferences [1]



Emergence of collective behaviors

- Anticipating trajectories and objectives
- Taking into account the consequences of others' actions on the environment

1-Interactions' Signatures

Emergence of interaction signatures:

- The signatures reflect the contexts affording interactions as observed through enacted interaction
- Multiple contexts with their respective probabilities can encode movements of mobile affordances



- Visual interactions also have signatures, that encode both transformations produced by agent's movements but also by affordances' movements

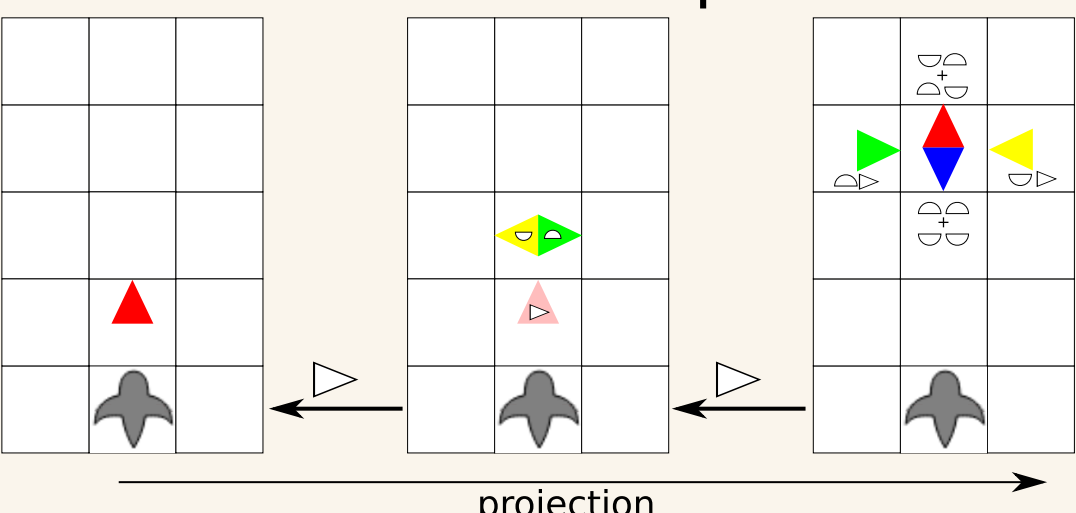
- These signatures also encode the orientation of mobile entities



2-Affordance recognition

Projection of interaction signatures:

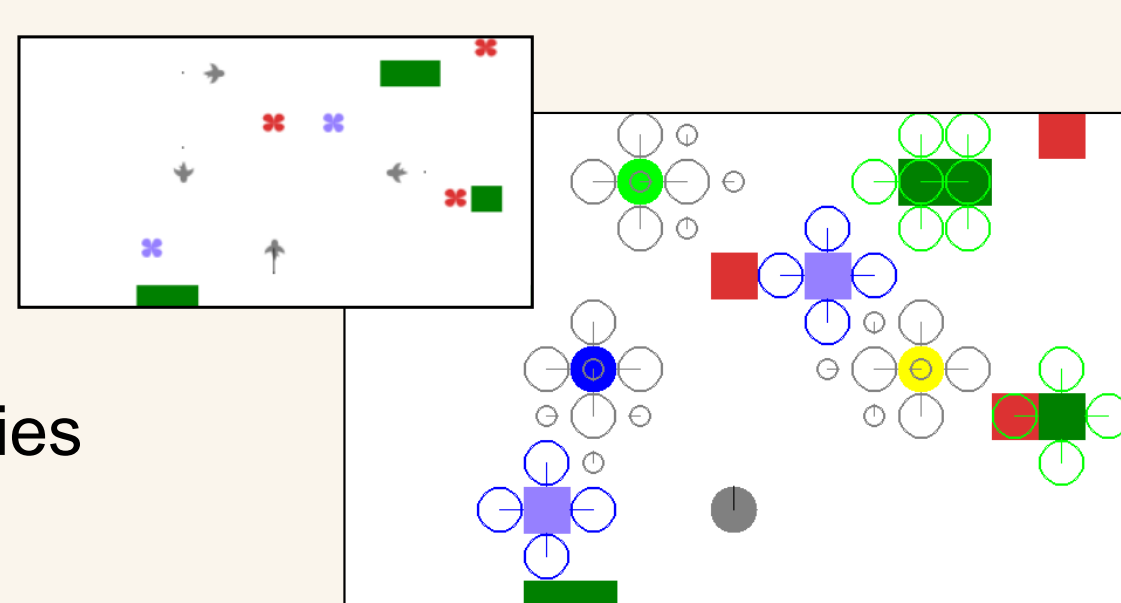
- A signature designates other interactions, that may have their own signatures, enabling their projection
- Mobile affordances have multiple contexts, but most of them lead to the position related to 'no movement'



- Localization through sequences of affordances

- Position of static objects

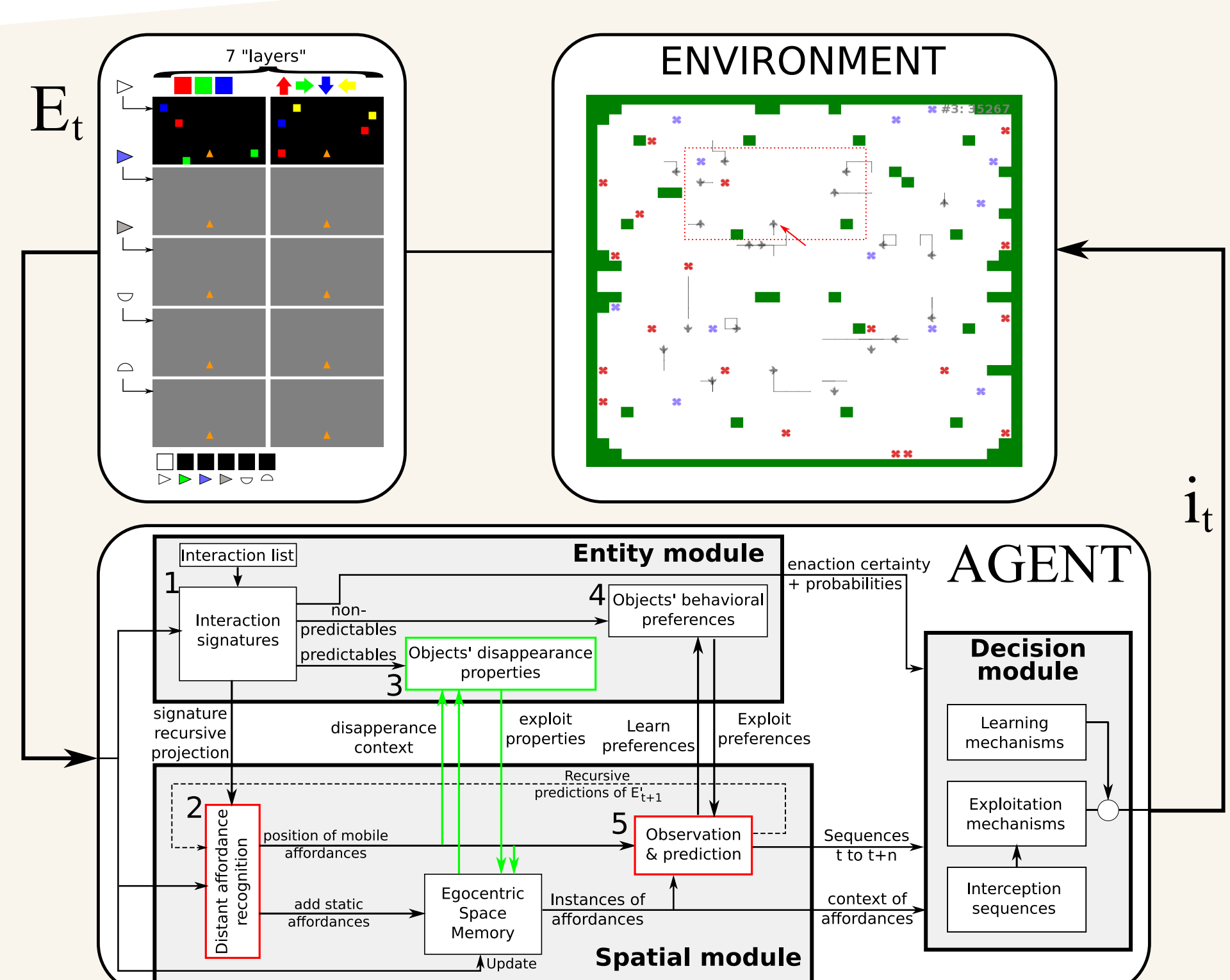
- Position and possible movements of mobile entities



Results

- Integration of mobile entities with an orientation
- Learn consequences of others' actions on the environment
- Bases for the emergence of collective/competitive behaviors constructed through sensorimotor scheme experience

Architecture

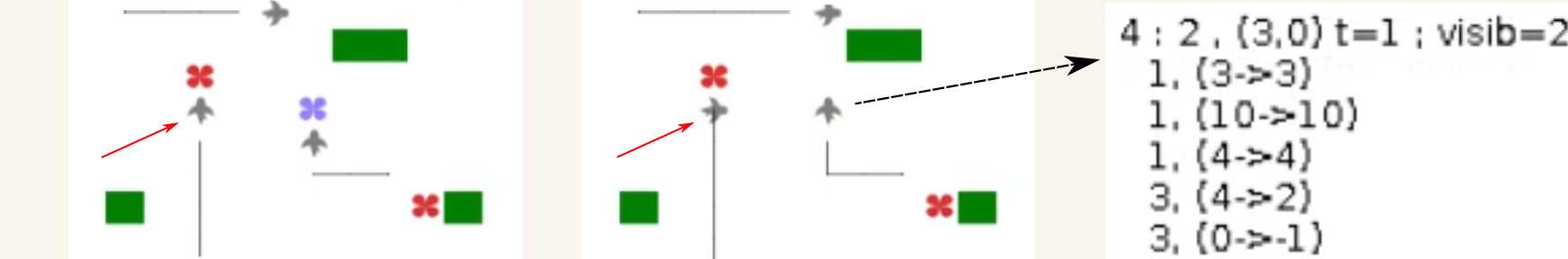


3-disappearance properties

Learning consequences of others' actions

- Another agent can add, remove or transform an affordance while interacting with it
- Differences may appear between observed context and space memory's content: a stored affordance may be absent
 - case 1: no known sequences (outside perception field)
 - case 2: affordance is hidden by another one
 - case 3: the affordance actually disappeared

- Principle: an affordance context centered on the missing affordance is recorded before and after its disappearance

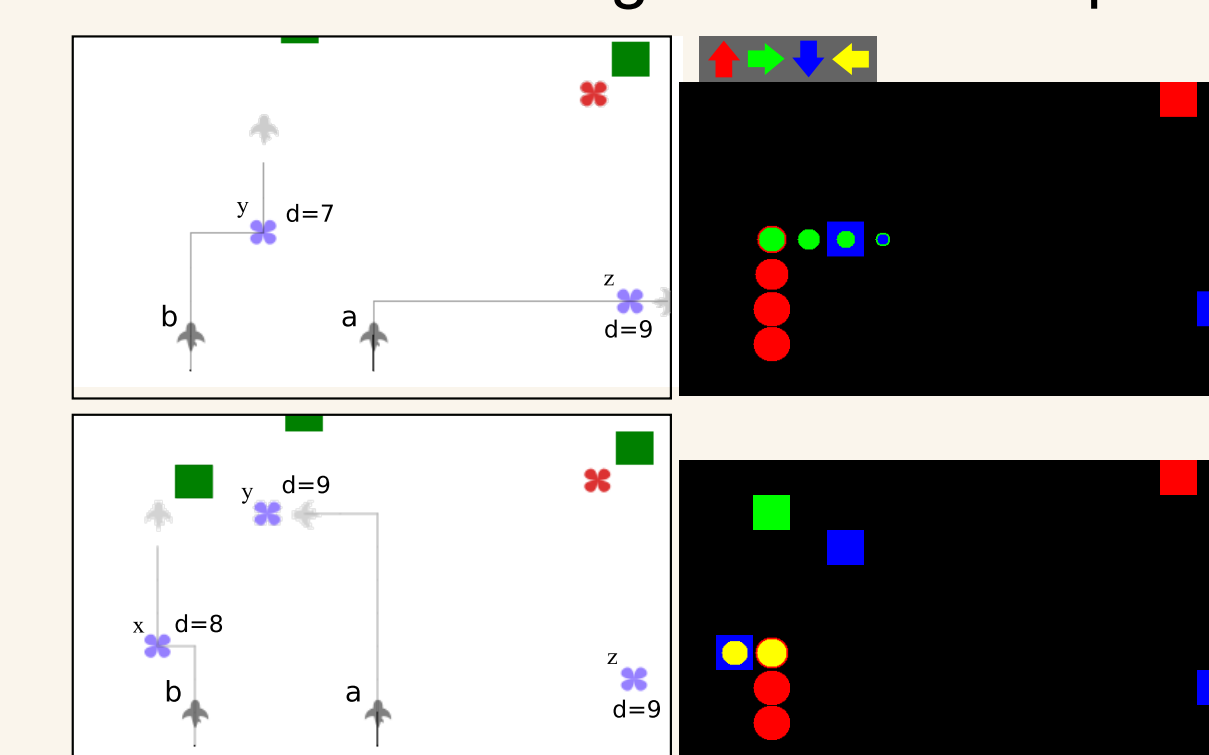


- The agent moves toward the affordance's position to confirm the disappearance
- The 'disappearance signatures' detect recurrent context elements that may be the cause of disappearance. For affordances of eat, (cuddle, 0, -1) is detected (other agent next to alga then at the same position)

5-Prediction and Decision

Generating collective behaviors

- Prediction of another agent's trajectory: the mechanism exploits 'next positions' and their associated utilities to recursively define a sequence of future positions
- When a disappearance signature is triggered, if the simulation's length is shorter than the affordance distance, it will be removed from the agent's decision process



- The agent ignores algae that it cannot reach before another agent eats them.

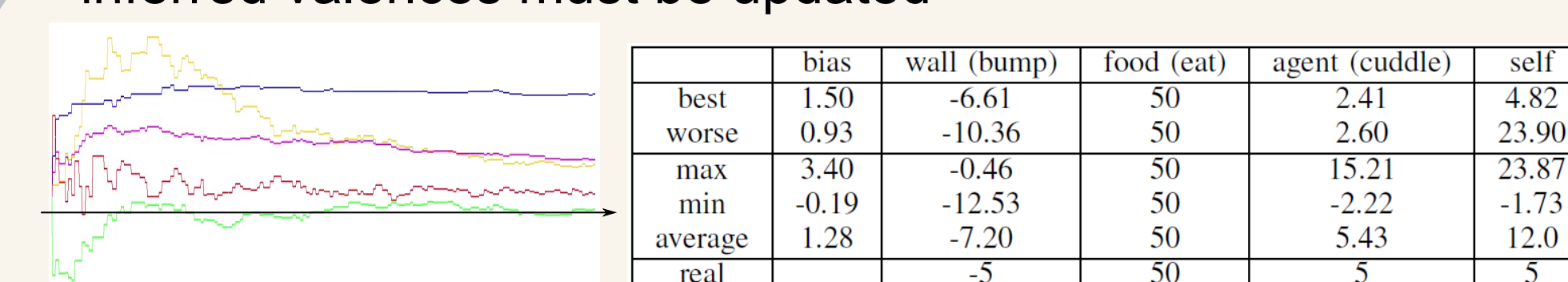
4-Behavioral Preferences

Inferring others' behavioral preferences

- RI agents behave according to surrounding affordances
- The agent considers itself as an affordance for others
- A utility value is computed from inferred valences

$$u_t^{\omega} = \sum_{(a_n, d_n) \in \bar{M}_t^{\omega}} v_{a_n}^{\omega} \times f(d_n)$$

- If utility value decreases over two consecutive steps, inferred valences must be updated



- Inferred valences of mobile entities show stronger variations between runs: individuation of initially identical agents?
- In Many runs, inferred valences for other agent and for self remain close: bases for self recognition as an agent?

Observations

- The agent considers possible evolutions of its environment
- The disappearance mechanism can be exploited to learn to predict visual occlusions
- Possible emergence of a strong individuation through different experiences of agent-agent interactions

Future work

- Emergence of collaborative behaviors
- Inferring hidden affordances through others' movements
- More complex sensorimotor capabilities
- Exploiting agents' individuation for behavior specialization

References

- [1] Interacting with Other Agents Without a-priori Knowledge: a Radical Interactionist Architecture Developing From the Ground-Up the Ability to Infer Motivations and Predict Behaviour, Gay S.L., Suro F., Jamont J.-P., ICDL 2025.
- [2] An affordance-based intersubjectivity mechanism to infer the behaviour of other agents, Gay S.L., Suro F., Jamont J.-P., ICDL 2023.
- [3] Artificial Interactionism: Avoiding Isolating Perception From Cognition in AI, Guillermin M., Georgeon O.L., Frontiers in Artificial Intelligence 2022.
- [4] Autonomous construction and exploitation of a spatial memory by a self-motivated agent, Gay S.L., Mille A. Georgeon O.L., Dutech A., Cognitive Systems Research, 2017